Sam Maione

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Education

Binghamton University — Binghamton, NY
B.Sc. in Integrative Neuroscience | Sociology (Minor) — 2019–2023

Experience

Laboratory Technician / Manager — Johns Hopkins University
Look, Infer, Understand (LIU) Lab // PI: Shari Liu, Ph.D. — July 2023–Present

- Foundational lab organization.
- As the first lab manager / technician of our lab, I created and documented our lab standard analysis pipelines for (1) fMRI analysis, and (2) looking time analysis, along with writing and maintaining the corresponding umbrella IRB protocols.
- Administration and technical resources.
- On the administrative side, I managed the budget, website, events, inventory, and reimbursements for all lab members. On the data management side, I wrote and maintained the lab manual, project board, and lab-wide backup server. On the technical side, I provided programming assistance and error debugging for over 25 studies.
- Recruitment coordination.
- After taking over the responsibilities of our old research coordinator, I led a partnership with Port Discovery, a local children's museum, where we conduct experiments and recruit for future studies.
- Independent cognitive neuroscience research.
- Alongside with Dr. Shari Liu, I conducted two novel fMRI studies on (1) the relationship between the intuitive physics network and the multiple demand network in the brain, published in *Neuropsychologia*, and (2) the role of physical cognitive resources in social prediction.

Research Assistant — Binghamton University
Binghamton Baby Lab // PI: Peter Gerhardstein, Ph.D. — Jan 2022–June 2023

- Honors thesis and publication.
- I led a partnership with Dr. Victoria Pelak and Dr. Peter Gerhardstein where we assessed the validity of a survey used to characterize visual snow symptoms. This study was funded by the Visual Snow Initiative and published in *Frontiers of Neurology*. In doing so, I earned the Undergraduate Research Award and Steven W. Kovacs Memorial Award for my commitment to psychological research.
- Developmental study support.
- I conducted in-person eye tracking and EEG studies with children aged 3-9. I also assisted in programming these experiments, and presented these projects at university research conferences.

Publications

1. **Maione, S. M.**, & Liu, S. (2026). Distinct and shared neural resources for dynamic object processing and spatial working memory. *Neuropsychologia, 232,* 109561. [OSF] [Paper]
2. Kim, M., **Maione, S. M.**, & Liu, S. (In review). Shared neural representations of barriers to agents' actions and objects' motions. [Pre-print]
3. **Maione, S. M.**, Pelak, V.S., & Gerhardstein, P. (2025). Assessment of a Novel Patient Reported Outcome Measure for Visual Snow Syndrome: The Colorado Visual Snow Survey 2.0. *Frontiers in Neurology, 16,* 1664310. [OSF] [Paper]

Conference presentations

1. **Maione, S. M.**, Drissi, A., Liu, K., Kim, M., & Liu, S. (2026). Social objects: Tracking agents relies on neural resources associated with dynamic social and physical processing. *Society for Philosophy and Psychology*, Baltimore, MD. [Poster]
2. **Maione, S. M.**, & Liu, S. (2024). Frontoparietal regions engaged in physical prediction are also involved in spatial working memory. *Conference on Cognitive Computational Neuroscience*, Boston, MA. [Conference abstract] [Poster]
3. Kim, M., **Maione, S. M.**, Drissi, A., & Liu, S. (2024). Cortical regions preferentially engaged during social and physical processing represent obstacles to agent action and object motion. *Conference on Cognitive Computational Neuroscience*, Boston, MA. [Conference abstract]
4. **Maione, S. M.**, & Gerhardstein, P. (2023). Perception through Visual Snow Syndrome: Faces, Scenes and Self-Reported Symptoms. *Binghamton University Psi Chi Fair*, Binghamton, NY. **(Undergraduate Research Award)**
5. **Maione, S. M.**, Duggan, N., & Gerhardstein, P. (2022). Changing Orientation Anisotropy Influenced by the Digital Age. *Binghamton University Psi Chi Fair*, Binghamton, NY.

Skill highlights

- **Cognitive Neuro.:** fMRI, TMS/rTMS, EEG
- **Dev. Psychology:** looking time, eye tracking
- **Data collection:** in-person, Connect, SONA
- **Software:** Google/Microsoft/Adobe suites

Programming highlights

- **R:** RMarkdown, tidyverse, lme4, ggplot2
- **Python:** Jupyter, scikit-learn, nipype/nibabel
- **Shell:** SLURM, unix cluster management
- **Other:** Psychtoolbox-3, Singularity, Docker

Scholarships and Awards

- Steven W. Kovacs Memorial Award [Link] 2023
- Binghamton University Undergraduate Research Award [Link] 2023
- Dr. Dominick A. and Susan G. Artuso Scholarships 2022

Other work experience

- Emerging Leaders Program Mentor 2022–2023
  I provided a 90-minute, structured, weekly space for 13 newly matriculated students to develop professional skills and become acclimated to college life. Through this experience, I mentored these students through a partnership with the Binghamton Rescue Mission, a homeless shelter, where we raised over $500 for new resident supplies.
- Supplemental Instruction Leader 2021–2023
  I held 60-minute weekly review sessions for an introductory psychology lecture, with attendance ranging from 20-135 students per session. Alongside review sessions, I helped students draft cover letters for campus internships and research opportunities.
- Pharmacy Technician 2017–2023
  I was a pharmacy technician and cashier throughout high school and college to pay for my educational expenses. Working in retail taught me a surprising amount of interpersonal skills to calmly and efficiently navigate high stress situations.